

Exercise 1-5

Perform the following conversions.

- a) 3.2 g Cu =mol Cu
(Cu = 64 g/mol)
- b) 0.2 mol Ag=.....g Ag
(Ag = 108g/mol)
- c) 6.4 g S =.....mol S =.....S atoms
(S = 32 g/mol)

Exercise 1-6

- a) What is the mass of 0.04 mole of N_2 ?
- b) What is the number of moles in 5.6 g of PCl_5 ?

$$M(NH_3) = (1 \times 14) + (3 \times 1) = 17 \text{ g/mol}$$

$$n(\text{mol}) = \frac{m(\text{g})}{M(\text{g/mol})} = \frac{85(\text{g})}{17(\text{g/mol})} = 5 \text{ mol } NH_3$$

Example 1-9

Calculate the mass of 0.7 mol of manganese dioxide (MnO_2).

Solution:

$$M(MnO_2) = (1 \times 55) + (2 \times 16) = 87 \text{ g/mol}$$

$$m(\text{g}) = n(\text{mol}) \times M(\text{g/mol})$$

$$m(\text{g}) = 0.7(\text{mol}) \times 87(\text{g/mol}) = 60.9 \text{ g } MnO_2$$

1-14-2-Applications of Mole Concept

As we have told before, the mass of 1 mole of carbon atom is 12 grams. According to this, it is possible to calculate the mass of 1 carbon atom.

$$= \frac{12 \text{ g}}{\text{Avogadro's number}} = \frac{12 \text{ g}}{6.023 \times 10^{23}} = 1.995 \times 10^{-23} \text{ g/atom}$$

We can write the following equation:

$$\text{Number of moles} = \frac{\text{Number of particles (atoms, ions or molecules)}}{\text{Avogadro's number}}$$

Example 1-10

- a) Calculate the number of moles of 3.01×10^{25} water molecules.
- b) Calculate the number of molecules in 0.02 mole of CO_2 .

Solution:

$$\text{a) Number of moles} = \frac{\text{number of molecules}}{\text{Avogadro's number } (N_A)}$$

$$= \frac{3.01 \times 10^{25}}{6.023 \times 10^{23}} = 50 \text{ moles } H_2O$$

$$\text{b) Number of molecules} = \text{number of moles} \times \text{Avogadro's number}$$

$$= 0.02 \times 6.023 \times 10^{23} = 1.2 \times 10^{22} \text{ } CO_2 \text{ molecules}$$

Example 1-15

In 10 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$,

- a) Find out mass of Cu.
- b) Find out mass percentage of crystal water.

Solution:

a) Molar mass of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$:

$$M(\text{CuSO}_4 \cdot 5\text{H}_2\text{O}) = 1 \times 64 + 1 \times 32 + 4 \times 16 + 5(2 \times 1 + 1 \times 16) \\ = 250 \text{ g/mol}$$

$$\text{Mass of Cu} = \left(\frac{64}{250} \right) \times 10 = 2.56 \text{ g}$$

b) Using the same equation;

$$\text{Mass of crystal water} = \left(\frac{18 \times 5}{250} \right) \times 10 = 3.6 \text{ g}$$

1-12-CHEMICAL FORMULA

The chemical composition of compounds represented with formulas, which are its constituent elements symbols set with the number of atoms of those elements in the molecule and the composition of known chemical matter can be expressed in different formats including:

1-12-1 Empirical Formula

It is the simplest formula that gives the minimum limit of absolute information about the compound, it sets the rational number of the atoms included in the compound. For example, one molecule of benzene consist of 6 carbon atoms and 6 hydrogen atoms, so the greatest common divisor for these numbers is 6, and by dividing the number of atoms over 6 we will get the empirical formula for benzene which is CH.

The same for the water molecule which is consist of two atoms of hydrogen and one atom of oxygen, so the empirical formula will be H_2O , and so for one molecule of glucose which consist of 6 carbon atoms, 12 hydrogen atoms and 6 oxygen atoms, so the greatest common division for these numbers is 6, and by dividing he atom numbers by 6 the empirical formula will be CH_2O .

How to find the empirical formula for compounds?

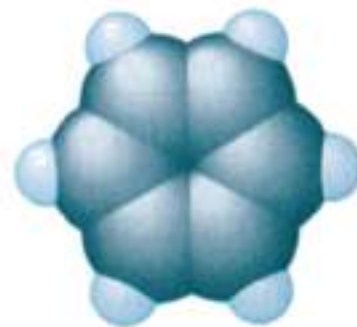
We should follow the steps below to find an empirical formula of a compound:

- a) The elements forming a compound are analyzed.
- b) The ratio of number of atoms is found by dividing mass of element or its percent ratio with atomic mass.

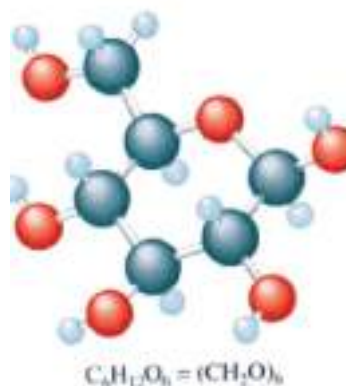
$$\text{Ratio of no. atoms of element} = \frac{\text{Mass of element or percent ratio}}{\text{Atomic mass}}$$

Exercise 1-11

Calculate the mass of Na in 25 g of $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$.



Benzene molecule



Glucose

Example 2-3

What is the equivalent of 688 Torr pressure in atm?

Solution:

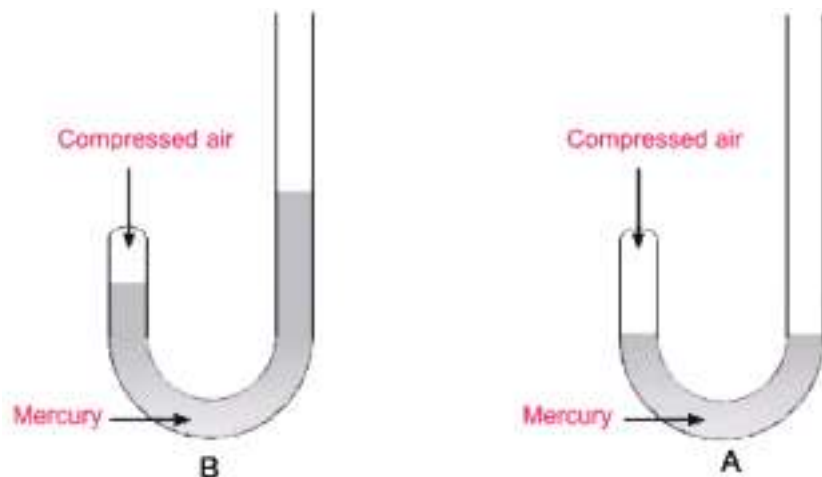
$$P (\text{atm}) = P \text{ Torr} \times \frac{1 \text{ atm}}{760 \text{ Torr}} = 688 \cancel{\text{Torr}} \times \frac{1 \text{ atm}}{760 \cancel{\text{Torr}}}$$

$$P \text{ atm} = 0.905 \text{ atm}$$

2-5-THE GAS LAWS

2-5-1-Volume-Pressure Relationship (Boyle's Law)

British scientist Robert Boyle made the first experiment on pressure and volume relationship. for gases He used a U-shaped glass tube with one arm shorter. The short arm is closed and filled with some gases and when mercury is added from the open and long arm, it starts to apply pressure on the gas. Until the gas is compressed to some volume, the amount of mercury increases and as shown in Fig. 2-1, the volume of gas has decreased.



Boyle observed that as pressure increased, volume of air decreased at constant temperature and the gas amount. Thus, Boyle's Law is stated "as long as temperature and amount of gas are kept constant, volume of gas is inversely proportional to pressure applied to it." This relationship is shown mathematically as follows:

$$V \propto \frac{1}{P} \quad \text{and} \quad V = k \frac{1}{P}$$

That means the product of pressure and volume is equal to a constant. According to this law, at constant temperature, for a certain amount of sample gas under P_1 pressure and V_1 volume is taken and if we would like to change P_2 pressure or V_2 volume of this gas, we can use the following equation:

$$P V = k \text{ (k is constant)}$$

Do you know that?



Air bags are actually inflated by the equivalent of a solid rocket booster. Sodium azide (NaN_3) and potassium nitrate (KNO_3) react very quickly to produce a large pulse of hot nitrogen gas. This gas inflates the bag, which literally bursts out of the steering wheel or dashboard as it expands.

Figure-2-1

- A) Pay attention to the air when some mercury is added.
B) We can observe that some more mercury is added, volume of air reduced to half

Do you know that?

The pressure on us is the pressure of air surrounding the Earth. The thickness of the atmosphere is 500 miles. In other words, we live under the ocean of atmosphere.

Exercise 2-4

If the pressure of 50 L of gas in a balloon was reduced from 1 atm to 0.9 atm, what will be the final volume of the gas?

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \text{ or } \frac{P_1}{P_2} = \frac{T_1}{T_2}$$

(Volume and amount of gas constant)

Example 2-7

Why shouldn't deodorant and spray cans be disposed of in fire? We can understand the reason better with the following question.

The pressure of a spray can is 3 atm at 17°C. What will be its pressure at 187°C?

Solution:

When temperature increases, kinetic energy of molecules in the can increases. Therefore, number of collisions of gas molecules with the walls of the can and also pressure increase. This causes can to burst.

The following steps are applied for solution:

We convert °C to K.

$$T \text{ (K)} = t \text{ (°C)} + 273$$

$$T_1 \text{ (K)} = 17 + 273 = 290 \text{ K} \quad P_1 = 3 \text{ atm}$$

$$T_2 \text{ (K)} = 187 + 273 = 460 \text{ K} \quad P_2 = ? \text{ atm}$$

We find out P_2 through Gay-Lussac Law.

$$\frac{P_1}{T_1} = \frac{P_2}{T_2} \quad T_1, P_1, T_2 \longrightarrow P_2$$

$$P_2 = \frac{P_1 T_2}{T_1} = \frac{(3 \text{ atm})(460 \text{ K})}{(290 \text{ K})}$$

$$P_2 = 4.75 \text{ atm}$$

2-6-THE COMBINED GAS LAW

If we re-write mathematical expressions of gas laws;

Boyle's Law $P \cdot V = k$

Charles' Law $\frac{V}{T} = k'$

Gay-Lussac's Law..... $\frac{P}{T} = k''$

A combined law can be written from those 3 laws:

$$\frac{PV}{T} = K \quad K(\text{Constant})$$

We can write the change in a gas with constant amount from P_1 , V_1 and T_1 conditions to P_2 , V_2 and T_2 conditions.

Exercise 2-5

1 L of CO_2 is found in a balloon at 27°C. What happens when we put the balloon in a pool at 270K?

Exercise 2-6

The air pressure of tires of a car is 1.8 atm at 20°C. The car's owner wants to go to Basra. When he arrives at Basra, what will be the air pressure of tires at 36°C?

For two different amounts of gases (n_1, n_2) and two different volumes (V_1, V_2) of them, we can use the following equation.

$$\frac{V_1}{n_1} = \frac{V_2}{n_2} \quad (\text{at constant temperature and pressure})$$

Avogadro's Law is expressed as "equal volumes of all gases, at the same temperature and pressure, have the same number of moles."

We can observe this in Fig 2-3.



Figure-2-3

When we withdraw gas from the environment, so, number of moles decreases. When we add some more gas, number of moles increases and gas volume increases.

2-7-1-MOLAR VOLUME

The value obtained by dividing volume to no. of moles is called as molar volume. As no. of moles is shown with (n) and volume with (V), V_m expresses molar volume.

$$V_m = \frac{V \text{ (L)}}{n \text{ (mol)}} = \text{L} / \text{mol}$$

Molar volume of a gas is equal to 22.4 L (22414cm³) at standard temperature and pressure (STP). (1atm = 760 Torr, 0°C = 273 K).

Molar mass (M) is found by dividing mass (m) to no. of moles (n) as we have studied in Chapter 1.0.0.

$$M = \frac{m \text{ (g)}}{n \text{ (mol)}} = \text{g} / \text{mol}$$

Example 2-9

If the volume of 1 mole of hydrogen gas is 22.4 L at standard temperature and pressure, what is the volume of 3 moles of hydrogen gas at the same conditions?

Solution:

$$\frac{V_1}{n_1} = \frac{V_2}{n_2} \quad \boxed{V_1, n_1, n_2} \rightarrow \boxed{V_2}$$

$$V_2 = \frac{V_1 n_2}{n_1} = \frac{(22.4 \text{ L})(3 \text{ mol})}{(1 \text{ mol})}$$

$$V_2 = 67.2 \text{ L}$$

Exercise 2-7

The volume of a sample of CO₂ is 4 L at 66°C temperature and 1.2 atm pressure. At 42°C, the volume becomes 1.7 L. Calculate the pressure at this temperature. (The no. of moles hasn't changed.)

Do you know that?

Gay-Lussac discovered the relationship between chemical reactions and volume. But other chemists didn't accept his theory then and they didn't know composition of substances. Therefore, chemical equations couldn't be written at that time. The studies of Gay-Lussac and Avogadro enabled writing of chemical equations.

2-8-IDEAL GAS LAW

We can obtain a single equation by combining 4 gas laws:

$$PV = k \quad \text{..... Boyle's Law}$$

$$\frac{V}{T} = k' \quad \text{..... Charles's Law}$$

$$\frac{P}{T} = k'' \quad \text{..... Gay-Lussac's Law}$$

$$\frac{V}{n} = k''' \quad \text{..... Avogadro Law}$$

$$V \propto n \frac{T}{P}$$

We can get a single equation from the equations above:

When we convert the ratio to an equation:

$$V = (\text{constant}) \ n \frac{T}{P}$$

Exercise 2-8

Calculate the molar volume of 3 moles of a gas with a volume of 36 L.

By inserting gas constant (R), we get the final equation will be:

$$PV = nRT \quad \text{ideal gas law}$$

We can apply the equation above can be applied to gases which are applied 4 laws. This kind of gases are called as ideal gases. R is the general constant for ideal gases(R) (0.082). This equation must be used along with P(atm) , V (L), n (number of moles) and T(K) units in mathematical calculations.

In order to calculate R value, we take 1 mole of an ideal gas with 22.414 L volume at standard temperature and pressure (0°C and 1 atm).

$$PV = nRT$$

$$R = \frac{PV}{nT} = \frac{1 \text{ atm} \times 22.414 \text{ L}}{1 \text{ mol} \times 273 \text{ K}}$$

$$R = 0.082 \frac{\text{atm} \cdot \text{L}}{\text{mol} \cdot \text{K}}$$

In order to find R value in International System of Units, we take pressure as 101325 Pa (Pascal), volume as $22.4 \times 10^{-3} \text{ m}^3$, no. of moles as 1 mole and temperature as 273K.

$$PV = nRT$$

$$R = \frac{PV}{nT} = \frac{101325 \text{ Pa} \times 22.4 \times 10^{-3} \text{ m}^3}{1 \text{ mol} \times 273 \text{ K}}$$

$$R = 8.314 \text{ Pa} \cdot \text{m}^3 / \text{mol} \cdot \text{K}$$

Exercise 2-9

If 0.5 mole of a gas has 11.2 L of volume at standard temperature and pressure (STP), what is the no. of mole when it has a volume of 16.8 L?

$$PV = \left(\frac{m}{M}\right)RT \quad \dots\dots\dots (2)$$

or;

$$P M = \left(\frac{m}{V}\right)RT \quad \dots\dots\dots (3)$$

$$\rho = \frac{m \text{ (g)}}{V \text{ (L)}} \quad \text{from density formula}$$

ρ shows density and by using it in (3)

We get

$$P M = \rho RT \quad \dots\dots\dots (4)$$

By arranging Equation (4), we can calculate density of a gas with known molar mass and pressure at a certain temperature.

$$\rho = \frac{PM}{RT}$$

We can find out the mass of gas or molar mass from Equation (3):

$$P M = \left(\frac{m}{V}\right)RT$$

By re-arranging

$$m = \frac{PMV}{RT} \quad \text{mass of gas}$$

Example 2-11

Hydrazine (N_2H_4) is used as rocket propellant. Calculate the density of this substance at standard temperature and pressure (STP).

Solution:

$$\rho = \frac{PM}{RT} \quad \boxed{P, T, M} \longrightarrow \boxed{\rho}$$

Molar mass of hydrazine

$$M (N_2H_4) = (2 \times 14) + (4 \times 1) = 32 \text{ g/mol}$$

At standard conditions, the pressure is 1 atm and the temperature is 273K

Exercise 2-10

Calculate the number of moles of 10 L of oxygen under standard conditions.

$$m = \frac{MPV}{RT} \quad P, V, T, M \rightarrow m$$

$$m = \frac{44 \text{ g/mol} \times 5.46 \text{ atm} \times 3 \text{ L}}{0.082 \text{ atm L / mol K} \times 300 \text{ K}}$$

$$m = 29.3 \text{ g}$$

3. Calculate the mole number;

$$n = \frac{m \text{ (g)}}{M \text{ (g/mol)}}$$

$$n = \frac{29.3 \text{ g}}{44 \text{ g/mol}} = 0.67 \text{ mol}$$

Exercise 2-11

Calculate the density of oxygen gas (O_2) at 373K temperature and 5 atm pressure in g/L unit.

Exercise 2-12

Methane is a gas obtained by oil refining. Find out the volume of 0.5 mole of a sample from methane gas at 27°C temperature and 3 atm pressure.

Example 2-14

At 227°C temperature and 748 Torr pressure, the mass of a gas is 0.6 g in a 500 mL container. Calculate the molar mass of the gas.

Solution:

The following steps are applied for solution:

1) We convert the volume from mL to L.

$$V \text{ (L)} = V \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 500 \text{ mL} \times \frac{1 \text{ L}}{1000 \text{ mL}} = 0.5 \text{ L}$$

We convert the temperature from °C to K.

$$T \text{ (K)} = t \text{ (°C)} + 273$$

$$T = 227 + 273 = 500 \text{ K}$$

2) We convert the pressure from Torr to atm.

$$P \text{ atm} = P \text{ Torr} \times \frac{1 \text{ atm}}{760 \text{ Torr}} = 748 \text{ Torr} \times \frac{1 \text{ atm}}{760 \text{ Torr}} = 0.984 \text{ atm}$$

3) From Equation $M = \frac{mRT}{PV}$

$$M = \frac{0.6 \text{ g} \times 0.082 \text{ atm L / mol K} \times 500 \text{ K}}{0.984 \text{ atm} \times 0.5 \text{ L}}$$

$$M = 50 \text{ g /mol}$$

Example 2-15

The molar mass of 0.31 g of a gas is 32 g/mol and it has a pressure of 1.17 atm. What is the temperature when this gas has a volume of 0.23 L?